

Ilya Jahed

Computer Engineering Student & Researcher

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Professional Summary

Computer Engineering undergraduate at IUST (GPA 18.85/20) and research assistant at the Scalable Systems Lab under Prof. Reza Entezari-Maleki, working on graph machine learning for network intrusion and anomaly detection. Research training covers paper analysis, modular reimplementations, and code auditing of self-supervised graph-based NIDS methods (Anomal-E, SL-GAD, NEGSC), with ongoing private extension work on NEGSC.

Education

Computer Engineering, B.Sc.

Sep 2023 - Jun 2028

Iran University of Science and Technology (IUST) | Tehran, Iran | GPA: 18.85 / 20.00

- GPA: 18.85 / 20.00 (current, 7th semester)
- Focus: graph machine learning and anomaly detection for network security

Research Experience

Research Assistant

Jun 2026 - present

Scalable Systems Lab, IUST | Tehran, Iran

Research assistantship under Prof. Reza Entezari-Maleki focused on graph machine learning for network security. Progression: analysis and modular reimplementations of Anomal-E (including auditing preprocessing and target-encoding concerns); study and reimplementations of SL-GAD as a node-centric graph anomaly-detection baseline; audit and modular reimplementations of NEGSC. The active NEGSC extension work is private and incomplete.

- Analysed and modularised an Anomal-E reimplementations for self-supervised NIDS research (Jun-Jul 2026)
- Studied and reimplemented SL-GAD as a node-centric graph anomaly-detection baseline (Jul-Aug 2026)
- Audited and modularised NEGSC in a public reimplementations repository (Aug 2026)
- Developing additional NEGSC extensions privately; final method and results are not yet claimed (Aug 2026-present)

Teaching Assistant - Data Structures

Sep 2025 - Jan 2026

Iran University of Science and Technology (IUST) | Tehran, Iran

TA for the Data Structures course under Dr. Hossein Rahmani.

- Held office hours and graded assignments and exams
- Assisted students with algorithmic problem-solving and C++ implementation

Selected Projects

NEGSC Reimplementations and Audit

Aug 2026 - present

<https://github.com/Ilya-Jahed/negsc-fraud-detection>

Modular public reimplementations and code audit of NEGSC (2024), a self-supervised graph-contrastive NIDS method (NEGAT encoder with Wasserstein / Gromov-Wasserstein losses) on NF-BoT-IoT-v2 NetFlow data. The associated extension work is private and incomplete; no final method or result is claimed.

- Analysed the paper and reference notebooks; reconstructed the pipeline as modular Python code
- Documented six paper-vs-code discrepancies (dead hyperparameters, untrained per-step edge embedder, ndata key mismatch, and others)
- Public foundation for ongoing private NIDS extension research

Anomal-E Analysis and Modular Reimplementations

Jun 2026 - Aug 2026

<https://github.com/Ilya-Jahed/Anomal-E-Implementation>

Modular implementation and audit of Anomal-E, an edge-centric self-supervised GNN method (E-GraphSAGE + DGI) for network anomaly detection on NetFlow data.

- Analysed the paper and restructured the implementation into modular components (data pipeline, graph builder, encoder, DGI, detectors, trainer)
- Investigated preprocessing and target-encoding leakage concerns during the audit

- Vectorised graph construction (NetworkX to DGL/NumPy) to eliminate OOM issues on full datasets

SL-GAD Reimplementation and Study

Jul 2026 - Aug 2026

<https://github.com/Ilya-Jahed/sl-gad-reimplementation>

Reimplementation of SL-GAD (TKDE 2021), a node-centric self-supervised graph anomaly-detection method combining generative attribute regression with multi-view contrastive learning.

- Implemented the full training and multi-round evaluation pipeline with a CLI
- Single-run AUC of 0.9167 on Cora, within the paper's reported range (not a full reproduction study)
- Studied contrastive/generative ideas relevant to the broader NIDS direction; recognised the node-centric vs flow-level distinction

Skills

Graph ML & NIDS Research (Strong): Graph Neural Networks, Self-supervised & Contrastive Learning, Graph Anomaly Detection, Network Intrusion Detection, Paper Analysis & Code Auditing

Programming (Python-focused): Python (proficient), C (coursework), C#.NET (working knowledge), Bash

ML Tooling (Working knowledge): PyTorch, DGL, PyTorch Geometric, scikit-learn, PyOD, Jupyter

Security Literature (Familiar): Provenance-based IDS (ORTHRUS, UNICORN - introductory study), OS audit primitives, NetFlow datasets (NF-BoT-IoT-v2, NF-CSE-CIC-IDS2018-v2)

Tools (Practical): Git, Linux, LaTeX, Reproducible experimentation

Languages

Azerbaijani Turkish Native speaker

Persian (Farsi) Fluent (second language)

English Professional working proficiency - IELTS Academic in preparation

Research Interests

Research

Graph Anomaly Detection, Self-supervised NIDS, Contrastive Learning, Reproducibility, Provenance Graphs